

Does model ablation reveal how choices were generated?

A The inference we test

REPLACE REWARDS AT EVALUATION

Original rewards



Donor rewards



Model

Same choices · fixed model parameters



NEXT-CHOICE PROBABILITIES



Choice C: 76% → 78%

A small prediction change



Little reward dependence in the generating process?

Does this inference follow?

B Known ground truth

Synthetic agents, specified rules

TASK A Four-armed restless bandit

Past trials

Next



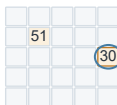
Mechanism A

Reward learning

Mechanism B

Choice kernel

TASK B Spatially correlated bandit



5 × 5 locations

Cue + selected feedback



Mechanism A

Generalize rewards



Mechanism B

Stay nearby

Reward weight shifts the balance

Choice-history-led

Reward-learning-led



Generate synthetic trajectories

C Validate the response

Fit models to synthetic trajectories



GRU



Transformer



LLaMA SFT



Symbolic models

RW / fitted generator

Neural: full input / choice-only fits

Apply the same replacement

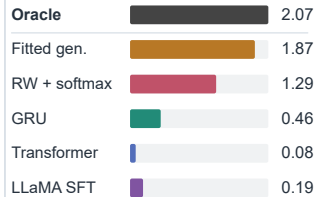


Choices held fixed · fitted parameters frozen



ONE GENERATOR · w = 1

Restless donor effect (nats / choice)



Same data, different ablation responses